

# A Scalable Gradient-Based Optimization Framework for Sparse Minimum-Variance Portfolio Selection

Presentation by  
Rohith Ramanan

# Minimum Variance Portfolio

Let the asset return vector be

$$x_\tau \in \mathbb{R}^p.$$

Portfolio weights satisfy

$$\beta \in \mathbb{R}^p, \quad \mathbf{1}^\top \beta = 1.$$

The portfolio return is  $\beta^\top x_\tau$ , and its variance is

$$\text{Var}(\beta^\top x_\tau) = \beta^\top \Sigma \beta.$$

**Minimum-variance portfolio**

$$\min_{\beta} \beta^\top \Sigma \beta \quad \text{s.t. } \mathbf{1}^\top \beta = 1$$

**Closed-form solution**

$$\hat{\beta} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

# Motivation for Sparse Portfolios

Classical minimum-variance portfolios are typically **dense**.

## Practical issues

- Very small weights across many assets
- High transaction costs
- Sensitivity to covariance estimation errors

**Goal:** select at most  $k$  assets.

## Sparse portfolio problem

$$\begin{aligned} \min_{\beta} \quad & \beta^{\top} \Sigma \beta \\ \text{s.t.} \quad & \mathbf{1}^{\top} \beta = 1, \quad \|\beta\|_0 \leq k \end{aligned}$$

This can be formulated as a **mixed-integer quadratic program** using the Big- $M$  formulation.

# Binary Formulation I

Introduce a binary selection vector

$$s \in \{0, 1\}^p$$

where  $s_j = 1$  if asset  $j$  is selected.

Let  $\Sigma[s]$  denote the principal submatrix corresponding to the selected assets.

The sparse portfolio problem can be written as

$$\min_{s \in \{0, 1\}^p} \min_{\beta[s] \in \mathbb{R}^{|s|}} \beta[s]^\top \Sigma[s] \beta[s]$$

subject to

$$\mathbf{1}^\top \beta = 1, \quad \mathbf{1}^\top s \leq k$$

# Binary Formulation II

For a fixed subset  $s$ , the inner minimization is the classical minimum-variance problem. Substituting back gives the discrete problem

$$\min_{s \in \{0,1\}^p} \frac{1}{\mathbf{1}^\top \Sigma^{-1} [s] \mathbf{1}} \quad \text{s.t. } \mathbf{1}^\top s \leq k,$$

or equivalently

$$\min_{s \in \{0,1\}^p} -\mathbf{1}^\top \Sigma^{-1} [s] \mathbf{1}. \quad \text{s.t. } \mathbf{1}^\top s \leq k$$

## Theorem 1

The optimal value of

$$\min_{s \in \{0,1\}^p} -\mathbf{1}^\top \Sigma^{-1} [s] \mathbf{1}, \quad \mathbf{1}^\top s \leq k$$

is non-increasing in  $k \in \{0, 1, \dots, p\}$ .

# Boolean Relaxation

Relax the binary variable

$$s \in \{0, 1\}^p$$

to a continuous vector

$$t \in [0, 1]^p.$$

Define

$$T_t = \text{Diag}(t), \quad \tilde{\Sigma}_t = T_t \Sigma T_t + \delta(I - T_t^2).$$

The **Boolean relaxation** of the discrete problem is

$$\min_{t \in C_k} f_\delta(t), \quad f_\delta(t) = -t^\top \tilde{\Sigma}_t^{-1} t.$$

**Constraint set**

$$C_k = \{t \in [0, 1]^p : \mathbf{1}^\top t \leq k\}.$$

# Properties of the Boolean Relaxation

## Theorem 2

The following properties hold for

$$f_\delta(t) = -t^\top \tilde{\Sigma}_t^{-1} t.$$

(i)  $f_\delta(t)$  is continuous on  $[0, 1]^p$ .

(ii) For every binary vector  $s \in \{0, 1\}^p$ ,

$$f_\delta(s) = -\mathbf{1}^\top \Sigma^{-1} [s] \mathbf{1} \quad \text{for all } \delta > 0.$$

(iii) For fixed  $t \in (0, 1)^p$ ,  $f_\delta(t)$  is monotonically increasing in  $\delta$ .

(iv) For any  $k$  and  $\delta > 0$ ,

$$\min_{t \in C_k} f_\delta(t) = \min_{t \in S_k} f_\delta(t),$$

where

$$S_k = \{t \in [0, 1]^p : \mathbf{1}^\top t = k\}.$$

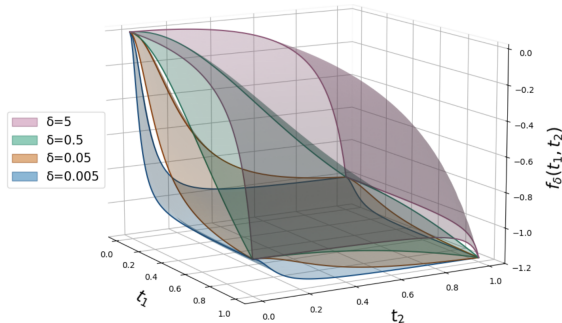
# Convex–Concave Transition

## Theorem 3

Let  $\eta_1$  and  $\eta_p$  be the largest and smallest eigenvalues of  $\Sigma$ .

- (i)  $f_\delta(t)$  is **strictly concave** on  $[0, 1]^p$  for  $\delta \geq \eta_1$ .
- (ii) For any  $\epsilon \in (0, 1)$ ,  $f_\delta(t)$  is **strictly convex** on  $[\epsilon, 1]^p$  for

$$\delta \leq \frac{3\eta_p\epsilon^2}{1 + 3\epsilon^2}.$$





# Continuity of the Relaxation

## Theorem 4

For each  $k$ , let  $h_k^*$  and  $D_k^*$  denote the optimal value and solution of the target problem. Define

$$h_k(\delta) = \min_{t \in C_k} f_\delta(t), \quad D_{k,\delta} = \arg \min_{t \in C_k} f_\delta(t).$$

For any  $\epsilon \in (0, \eta_1)$ ,

- $h_k(\delta)$  is continuous in  $\delta$ ,
- $D_{k,\delta}$  is compact-valued and upper hemicontinuous.

Moreover, if  $\delta_\ell \rightarrow \eta_1$  and  $t^{(\ell)} \in D_{k,\delta_\ell}$  with  $t^{(\ell)} \rightarrow t^*$ , then

$$t^* \in D_k^*.$$

## Implication

- Increase  $\delta \rightarrow \eta_1$
- Minimizers of  $f_\delta(t)$  converge to a solution of the original sparse portfolio problem.

# Algorithm

## Algorithm 1: Grid-FW( $\Sigma, \alpha, \epsilon, n, m$ )

- 1 Compute largest and smallest eigenvalues

$$\eta_1, \eta_p \text{ of } \Sigma$$

- 2 Initialize

$$t \leftarrow \frac{k}{p} \mathbf{1}, \quad s^* \leftarrow 0$$

- 3 Construct geometric grid

$$\delta_1, \dots, \delta_n \quad (\delta_n \rightarrow \eta_1)$$

- 4 For  $\ell = 1, \dots, n$

- Compute gradient  $\nabla f_{\delta_\ell}(t)$
- Select

$$s = \arg \min_{u \in \{0,1\}^p, \mathbf{1}^\top u = k} u^\top \nabla f_{\delta_\ell}(t)$$

- Update

$$t \leftarrow (1 - \alpha)t + \alpha s$$

- 5 Return best subset  $s^*$

# Practical Considerations

## Remark 1: Scaling

- In real datasets, eigenvalues of the covariance matrix  $\Sigma$  can be very small and span a wide range.
- This can cause numerical instability when choosing the parameter  $\delta$ .
- Solution: replace  $\Sigma$  with its **correlation matrix** before applying the Boolean relaxation.
- Accordingly,  $\eta_1$  and  $\eta_p$  are rescaled using

$$\min_j \Sigma_{jj}, \quad \max_j \Sigma_{jj}.$$

- This ensures the  $\delta$ -grid remains numerically stable.

## Remark 2: Running Time

- Algorithm performs  $nm$  iterations.
- Each step computes the gradient  $\nabla f_\delta(t)$ .
- This requires solving a linear system via conjugate gradient.
- Cost per iteration:  $O(p^2)$ .
- Total runtime:

$$O(nmp^2).$$

# Results

**Table:** Accuracy of our method compared to optimal portfolios for datasets with  $p = 31$  and  $p = 85$  assets. FP = number of false positives, Err = percentage variance error.

| $k$ | $p = 31$              |    |        |          | $p = 85$              |    |        |          |
|-----|-----------------------|----|--------|----------|-----------------------|----|--------|----------|
|     | Opt Var               | FP | Err(%) | Best Err | Opt Var               | FP | Err(%) | Best Err |
| 1   | $1.29 \times 10^{-3}$ | 0  | 0.0    | 0.0      | $4.50 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 2   | $7.99 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $2.74 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 3   | $7.15 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $2.19 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 4   | $6.75 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $1.97 \times 10^{-4}$ | 1  | 0.79   | 0.79     |
| 5   | $6.50 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $1.84 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 6   | $6.22 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $1.72 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 7   | $6.01 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $1.64 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 8   | $5.83 \times 10^{-4}$ | 1  | 0.26   | 0.26     | $1.56 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 9   | $5.66 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $1.52 \times 10^{-4}$ | 0  | 0.0    | 0.0      |
| 10  | $5.56 \times 10^{-4}$ | 0  | 0.0    | 0.0      | $1.47 \times 10^{-4}$ | 0  | 0.0    | 0.0      |

# Limitations

## Implementation gap

- Our implementation is written in Python.
- Competing solvers such as **CPLEX** are implemented in highly optimized low-level languages.
- Consequently, direct runtime comparisons are not fully favorable to our method.

## Theoretical vs practical guarantees

- The theory proves recovery of the exact sparse  $k$ -optimal solution as

$$\delta \rightarrow \eta_1$$

continuously.

- In practice:
  - $\delta$  is discretized over a grid.
  - Each step is solved approximately using gradient-based optimization.
- Therefore, exact optimality is not always guaranteed in implementations.

# References



S. Moka, M. Quiroz, V. Asimit, S. Muller. *A Scalable Gradient-Based Optimization Framework for Sparse Minimum-Variance Portfolio Selection*. arXiv:2505.10099, 2025.

## Proofs

# Proof of Theorem 1

Let

$$h(k) = \max_{s \in \{0,1\}^p, \mathbf{1}^\top s = k} \mathbf{1}^\top \Sigma_{[s]}^{-1} \mathbf{1}.$$

To show monotonicity, consider  $s'$  obtained from  $s$  by adding one asset.

WLOG let  $s'$  and  $s$  differ at the index  $p$ , then

$$\Sigma_{[s']} = \begin{bmatrix} \Sigma_{[s]} & a \\ a^\top & b \end{bmatrix}, \quad b > 0.$$

Using the Banachiewicz inversion formula,

$$\Sigma_{[s']}^{-1} = \begin{bmatrix} \Sigma_{[s]}^{-1} + \frac{1}{c} \Sigma_{[s]}^{-1} a a^\top \Sigma_{[s]}^{-1} & -\frac{1}{c} \Sigma_{[s]}^{-1} a \\ -\frac{1}{c} a^\top \Sigma_{[s]}^{-1} & \frac{1}{c} \end{bmatrix}$$

where

$$c = b - a^\top \Sigma_{[s]}^{-1} a.$$



# Proof of Theorem 1 (continued)

Compute

$$\mathbf{1}^\top \Sigma_{[s']}^{-1} \mathbf{1} = \mathbf{1}^\top \Sigma_{[s]}^{-1} \mathbf{1} + \frac{1}{c} (\mathbf{1}^\top \Sigma_{[s]}^{-1} a - 1)^2.$$

Hence

$$\mathbf{1}^\top \Sigma_{[s']}^{-1} \mathbf{1} \geq \mathbf{1}^\top \Sigma_{[s]}^{-1} \mathbf{1}$$

whenever

$$c = b - a^\top \Sigma_{[s]}^{-1} a > 0.$$

Since  $\Sigma_{[s']}$  is positive definite, its Schur complement satisfies  $c > 0$ .

Therefore

$$h(k) \leq h(k+1).$$

## Lemma 2

### Lemma 2

For every interior point  $t \in (0, 1)^p$ , the objective can be written as

$$f_\delta(t) = -\mathbf{1}^\top \Pi_t^{-1} \mathbf{1}$$

where

$$\Pi_t = \Sigma + \delta D_t, \quad D_t = T_t^{-2} - I.$$

Since  $t^\top = \mathbf{1}^\top T_t$ ,

$$f_\delta(t) = -\mathbf{1}^\top T_t \tilde{\Sigma}_t^{-1} T_t \mathbf{1} = -\mathbf{1}^\top (T_t^{-1} \tilde{\Sigma}_t T_t^{-1})^{-1} \mathbf{1} = -\mathbf{1}^\top (\Sigma + \delta(T_t^{-2} - I))^{-1} \mathbf{1}.$$

# Proof of Theorem 2

## Lemma 3

### Gradient and Hessian

For  $t \in (0, 1)^p$ ,

$$\nabla \hat{f}_\delta(t) = -2\delta \frac{v \odot (\hat{\Pi}_t^{-1} \mathbf{1})^2}{t^3}.$$

$$\hat{H}_\delta(t) = 2\delta \text{Diag}\left(\frac{v \odot (\hat{\Pi}_t^{-1} \mathbf{1})}{t^3}\right) \left[ 3\text{Diag}(t^2 \odot w^2) - 4\delta \hat{\Pi}_t^{-1} \right] \text{Diag}\left(\frac{v \odot (\hat{\Pi}_t^{-1} \mathbf{1})}{t^3}\right).$$

# Proof of Theorem 3